JUNIOR SCHOOL LEARNING HUB

GRADE 9 PRE-TECHNICAL STUDIES
END-OF-TERM 2 EXAMINATION

Name: _____	Adm. No.: _____	Date: _____	Class: _____
Time: 2 hours 30 minutes	Total: 100 marks	Term: 2	Learning Area: Pre-Technical Studies

INSTRUCTIONS TO LEARNERS

- Answer all questions in Sections A and B.
- Use appropriate technical terms and show calculations where required.
- Draw clear, labelled sketches where requested.
- Observe workshop safety precautions throughout the examination.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. Define the term design brief. (2 marks)

2. State two ways of identifying a need or problem in the community. (2 marks)

3. Give two qualities of a good design solution. (2 marks)

4. State two reasons for preparing a working drawing before making an artefact. (2 marks)

5. Name two common drawing instruments used in technical drawing. (2 marks)

6. State two safety rules to observe when using a hacksaw. (2 marks)

7. Name two personal protective equipment items used in a workshop. (2 marks)

8. State two properties of timber that make it useful in construction. (2 marks)

9. Give two examples of manufactured boards. (2 marks)

10. Name two common defects found in timber. (2 marks)

11. State two properties of metals that make them useful for making tools. (2 marks)

12. Give two examples of non-ferrous metals. (2 marks)

13. Name two methods of joining pieces of wood. (2 marks)

14. State two advantages of using screws instead of nails in some wood joints. (2 marks)

15. What is meant by the term circuit? (2 marks)

16. Name two components used to control or protect an electrical circuit. (2 marks)

17. State two uses of a simple electrical switch. (2 marks)

18. Give two examples of renewable energy sources that can be used in a home. (2 marks)

19. State two ways of caring for hand tools after use. (2 marks)

20. Give two reasons for evaluating a finished product. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Design process and technical drawing (10 marks)

a) Outline four stages followed when solving a design problem. (4 marks)

b) State three details that should appear on a working drawing. (3 marks)

c) Explain why dimensions must be accurate on a working drawing. (3 marks)

22. Workshop safety and tools (10 marks)

a) State four general workshop safety rules. (4 marks)

b) Describe the correct procedure for storing hand tools after use. (3 marks)

c) Give three possible consequences of unsafe workshop practices. (3 marks)

23. Materials and joining methods (10 marks)

a) Distinguish between hardwood and softwood. (2 marks)

b) State three factors considered when selecting materials for an artefact. (3 marks)

c) Describe two methods of joining timber. (4 marks)

d) State one reason for sanding a wooden surface before finishing. (1 mark)

24. Basic electricity and energy (10 marks)

a) Draw a simple circuit using a cell, switch and bulb, and label the components. (3 marks)

b) State two differences between a conductor and an insulator. (2 marks)

c) A 6 V battery is connected to a 3 Ω resistor. Calculate the current. (3 marks)

d) Give two ways of reducing electricity wastage at home or school. (2 marks)

25. Production, finishing and quality control (10 marks)

a) State three reasons for marking out materials before cutting. (3 marks)

b) Explain two purposes of applying a finish to a wooden or metal product. (4 marks)

c) State three checks carried out when evaluating a finished artefact. (3 marks)

26. Entrepreneurship and product costing (10 marks)

a) State three qualities of a successful small-scale entrepreneur. (3 marks)

b) A learner makes a stool using materials costing KSh 450 and spends KSh 150 on other production expenses. Calculate the total production cost. (2 marks)

c) If the stool is sold for KSh 850, calculate the profit. (2 marks)

d) Give three ways a learner can market a product made at school. (3 marks)

MARKING SCHEME

Grade 9 Pre-Technical Studies – End-of-Term 2 Examination

Total: 100 marks | Section A: 40 marks | Section B: 60 marks

SECTION A (40 MARKS)

1. A short statement describing a problem, need or task to be solved through design. (2)

2. Observation; interviews; questionnaires; community discussion; research. Any two. (2)

3. Functional, safe, affordable, durable, attractive, sustainable and fit for purpose. Any two. (2)

4. Communicates dimensions; guides manufacture; shows shape/details; reduces errors and wastage. Any two. (2)

5. Any two: ruler, set square, compass, protractor, drawing board, T-square, pencils. (2)

6. Secure workpiece; use correct blade tension; cut away from body where appropriate; keep hands clear; use steady strokes; do not use damaged blade. Any two. (2)

7. Any two: goggles, gloves appropriate to task, overalls/apron, safety boots, dust mask, ear protection. (2)

8. Any two: strength, durability, workability, availability, appearance, insulation. (2)

9. Any two: plywood, blockboard, particleboard/chipboard, MDF. (2)

10. Any two: knots, shakes, splits, warping, decay, insect attack. (2)

11. Any two: strength, toughness, malleability, ductility, conductivity, hardness, durability. (2)

12. Any two: aluminium, copper, zinc, brass, bronze, lead. (2)

13. Any two: nailing, screwing, gluing, dowelling, mortise-and-tenon, dovetail, halving. (2)

14. Any two: can be removed; provide firm grip; less likely to loosen under some conditions; allow dismantling/adjustment. (2)

15. A complete path through which electric current can flow. (2)

16. Any two: switch, fuse, circuit breaker, resistor, protective earth device. (2)

17. Any two: opens/closes a circuit; controls operation of a lamp/appliance; isolates power. (2)

18. Any two: solar, wind, biogas, small-scale hydro, geothermal where available. (2)

19. Any two: clean and dry; oil moving parts; sharpen where appropriate; store in correct place; inspect and repair. (2)

20. Any two: check function, safety, quality, accuracy, durability and whether it meets the design brief; identify improvements. (2)

SECTION B (60 MARKS)

21. Design process and technical drawing (10 marks)

Identify/investigate the problem; generate ideas; develop/select a solution and plan; make/implement; test/evaluate and improve. Award any four valid stages. (4)

Dimensions; title/name of artefact; scale; material; labels/notes; views/projection; date/drafter. Any three. (3)

They guide correct measurement and cutting; ensure parts fit; reduce material wastage and errors. (3)

22. Workshop safety and tools (10 marks)

Wear PPE; keep floor/workbench tidy; use correct tool; avoid horseplay; follow instructions; report damaged tools; keep guards in place. Any four. (4)

Clean and dry tools; inspect for damage; return to labelled storage; protect sharp edges; store safely. Award any three explained points. (3)

Cuts/burns; eye injuries; electric shock; fire; damage to tools/materials; poor-quality work. Any three. (3)

23. Materials and joining methods (10 marks)

Hardwood comes from broad-leaved/angiosperm trees; softwood generally comes from coniferous trees. Do not define solely by hardness. (2)

Strength; durability; cost; availability; appearance; workability; intended use; safety/environmental impact. Any three. (3)

Any two described: nailing, screwing, gluing, dowelling, halving, mortise-and-tenon, dovetail. Award 2 each for correct method and explanation. (4)

Removes roughness/splinters and prepares a smooth surface for finishing. (1)

24. Basic electricity and energy (10 marks)

Correct complete circuit (1); cell labelled (1); switch and bulb labelled (1). (3)

Conductors allow current to flow easily; insulators resist/prevent current flow. Give suitable examples or explanation. (2)

$I = V/R = 6/3 = 2$ A. Award formula (1), substitution (1), answer with unit (1). (3)

Switch off unused lights/appliances; use efficient lamps; unplug chargers; use daylight; maintain appliances. Any two. (2)

25. Production, finishing and quality control (10 marks)

Improves accuracy; guides cutting/drilling; reduces wastage; ensures correct dimensions and fit. Any three. (3)

Protects from moisture/corrosion/wear; improves appearance; makes cleaning easier; increases durability. Any two explained. (2 each)

Dimensions/accuracy; strength/stability; functionality; neatness/finish; safety; suitability to design brief. Any three. (3)

26. Entrepreneurship and product costing (10 marks)

Creativity; persistence; honesty; initiative; willingness to learn; planning; responsible risk-taking. Any three. (3)

450 + 150 = KSh 600. (2)

Profit = selling price − total cost = 850 − 600 = KSh 250. (2)

Display/exhibition; posters; word of mouth; school/community fair; online platforms with permission; direct selling. Any three. (3)

ADMINISTRATION NOTE

This is an original teacher-prepared assessment resource for practice and school-based assessment. Before administration, the Pre-Technical Studies teacher should confirm that the assessed design, materials, workshop, electrical and enterprise concepts match the actual Grade 9 Term 2 teaching sequence and the school scheme of work. Practical tasks must be supervised and conducted with appropriate safety controls.

Prepared by: Junior School Learning Hub